Schemas and formats

Data engineering · Level 4 · Lesson 1



In this lesson you will

- Choose a schema: normalised for apps, a star for reports.
- Compare row and column formats and compression.
- Parse and type a CSV, counting rows that break the schema.

Key ideas

- Normalise for writes; star for reads.
- JSON repeats names; columns squeeze out repeats.
- Count what doesn’t fit, with a reason.

Size it on paper

1. Write (4 min): three records, two ways
2. Scale (4 min): a term, a district
3. Star (6 min): sketch the schema
4. Decide (3 min): which format?

Use made-up records. The point is the ratio between formats, not the exact bytes.

Let’s discuss

1. Why does a star schema keep names out of the fact table?
2. When would you still choose JSON?
3. What should happen to a row that breaks the schema?

Exit ticket

- Name one table in a star schema for school attendance, and whether it’s a fact or a dimension.